PGCP-AC AUGUST 26
Assignment No- 2

1. Write a SQL query to create Employee & Department table with specific columns and constraints.(**Note**- Create Table, insert data as per your Choice)

Answer:

```
mysql> CREATE TABLE DEPARTMENT (DEPT_ID INT PRIMARY KEY,DEPT_NAME VARCHAR(30),LOCATION VARCH
AR(30));
Query OK, 0 rows affected (0.03 sec)

mysql> CREATE TABLE EMPLOYEE (EMP_ID INT PRIMARY KEY,NAME VARCHAR(30),SALARY DECIMAL(10,2),J
OINING YEAR,DEPT_ID INT,CONSTRAINT FK_DEPTID FOREIGN KEY(DEPT_ID) REFERENCES DEPARTMENT(DEPT
_ID));
Query OK, 0 rows affected (0.06 sec)
```

2. Write a SQL query to add any new column to an existing table.

Answer:

```
mysql> ALTER TABLE EMPLOYEE ADD COLUMN NOTE VARCHAR(20);
Query OK, 0 rows affected (0.03 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

3. Write a SQL query to insert multiple records (At least 5) into a table in a single operation.

Answer:

```
mysql> INSERT INTO EMPLOYEE VALUES (111,"SAIRAJ",30000,"2000",2,"JUNIOR ENGINEER"),
    -> (112,"ANIKET",50000,"2021",3,"ANALYST"),
    -> (113,"SUJATA",25000,"2018",5,"ANALYST"),
    -> (114,"SUJAL",45000,"2014",2,"HR"),
    -> (115,"MUKESH",30000,"2022",1,"HR");
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

4. Write a SQL query to delete all records from a table.

Answer:

```
mysql> DELETE FROM EMPLOYEE;
Query OK, 15 rows affected (0.02 sec)
```

5. Write a SQL query to update salary records from a table where id = 5.

Answer:

```
mysql> UPDATE EMPLOYEE SET SALARY = 20000 WHERE EMP_ID = 105;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

6. Write a query to grant SELECT privilege on the Emp table to user readonly_user.

Answer:

```
mysql> CREATE USER 'readonly_user'@'localhost' IDENTIFIED BY 'pass@123';
Query OK, 0 rows affected (0.02 sec)

mysql> GRANT SELECT ON TEMP.EMP TO 'readonly_user'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql>
```

7. Write a query to revoke INSERT privilege on the Emp table from user temp_user.

Answer:

```
mysql> REVOKE INSERT ON TEMP.EMP FROM 'temp_user'@'localhost';
Query OK, 0 rows affected (0.02 sec)
```

8. Write a query to insert a new employee into the Emp table and commit the transaction.

Answer:

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO EMP VALUES (111,"SAKSHI",30000,'2019',5,"DEVELOPER");
Query OK, 1 row affected (0.00 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.02 sec)
```

9. Write a query to update the Salary of an employee with EmpID = 101 and commit the transaction.

Answer:

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE EMP SET SALARY = 20000 WHERE EMP_ID = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)
```

10. Write a query to set a SAVEPOINT before updating the DeptID of an employee.

Answer:

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> SAVEPOINT S1;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE EMP SET DEPT_ID = 5 WHERE EMP_ID = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

11. Write a query to rollback to the previously created SAVEPOINT.

Answer:

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> SAVEPOINT S1;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE EMP SET DEPT_ID = 5 WHERE EMP_ID = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> ROLLBACK TO S1;
Query OK, 0 rows affected (0.02 sec)
```

12. Write a query to truncate all data from the Emp table.

Answer:

```
mysql> TRUNCATE TABLE EMP;
Query OK, 0 rows affected (0.09 sec)
```

13. Write a query to list all employees whose salary is greater than 50,000.

Answer:

```
mysql> SELECT * FROM EMP WHERE SALARY >= 50000;
+--------+--------+----------+---------+---------+------+
| EMP_ID | NAME   | SALARY   | JOINING | DEPT_ID | NOTE |
+--------+--------+----------+---------+---------+------+
|    102 | Priya  | 52000.00 |    2021 |       2 | NULL |
|    104 | Sneha  | 60000.00 |    2020 |       1 | NULL |
|    106 | Neha   | 55000.00 |    2021 |       5 | NULL |
|    108 | Pooja  | 65000.00 |    2019 |       3 | NULL |
|    110 | Anjali | 58000.00 |    2020 |       5 | NULL |
+--------+--------+----------+---------+---------+------+
5 rows in set (0.00 sec)
```

14. Write a query to increase the salary of all employees in DeptID = 5 by 10%.

Answer:

```
mysql> UPDATE EMP SET SALARY = SALARY+(SALARY*0.1) WHERE DEPT_ID = 5;
Query OK, 2 rows affected (0.02 sec)
Rows matched: 2  Changed: 2  Warnings: 0
```

15. Write a query to count the total number of employees in each department.

Answer:

```
mysql> SELECT DP.DEPT_NAME,COUNT(E.EMP_ID) AS TOTAL_EMP FROM DEPARTMENT DP JOIN EMP E ON DP.
DEPT_ID = E.DEPT_ID GROUP BY DP.DEPT_NAME;
+-----------+-----------+
| DEPT_NAME | TOTAL_EMP |
+-----------+-----------+
| IT        |         2 |
| HR        |         2 |
| Finance   |         2 |
| Marketing |         2 |
| Sales     |         2 |
+-----------+-----------+
5 rows in set (0.00 sec)
```

16. Write a query to list departments that have no employees. & create a view for those who belong to the city of Mumbai.

Answer:

```
mysql> SELECT DP.DEPT_NAME FROM DEPT DP LEFT JOIN EMP E ON DP.DEPT_ID = E.DEPT_ID WHERE E.DE
PT_ID IS NULL;
+-----------+
| DEPT_NAME |
+-----------+
| Sales     |
+-----------+
1 row in set (0.00 sec)

mysql> CREATE VIEW MUMBAI_EMP AS SELECT E.NAME FROM EMP E JOIN DEPT DP ON E.DEPT_ID = DP.DEP
T_ID WHERE DP.LOCATION = 'MUMBAI';
Query OK, 0 rows affected (0.03 sec)

mysql> SELECT * FROM MUMBAI_EMP;
+--------+
| NAME   |
+--------+
| Neha   |
| Anjali |
+--------+
2 rows in set (0.00 sec)
```

17. Write a query to display the highest salary among all employees.

Answer:

```
mysql> SELECT SALARY FROM EMP ORDER BY SALARY DESC LIMIT 1 ;
+----------+
| SALARY   |
+----------+
| 65000.00 |
+----------+
1 row in set (0.00 sec)
```

18. Write a query to display the 2nd highest salary among all employees.

Answer:

```
mysql> SELECT SALARY FROM EMP ORDER BY SALARY DESC LIMIT 1 OFFSET 1;
+----------+
| SALARY   |
+----------+
| 63800.00 |
+----------+
1 row in set (0.00 sec)
```

19. Write a query to show all employees ordered by JoiningDate in descending order.

Answer:

```
mysql> SELECT * FROM EMP ORDER BY JOINING DESC;
+--------+--------+----------+---------+---------+------+
| EMP_ID | NAME   | SALARY   | JOINING | DEPT_ID | NOTE |
+--------+--------+----------+---------+---------+------+
|    109 | Vikas  | 35000.00 |    2024 |       4 | NULL |
|    103 | Amit   | 38000.00 |    2023 |       3 | NULL |
|    107 | Karan  | 48000.00 |    2023 |       2 | NULL |
|    101 | Rahul  | 45000.00 |    2022 |       2 | NULL |
|    105 | Rohan  | 42000.00 |    2022 |       4 | NULL |
|    102 | Priya  | 52000.00 |    2021 |       2 | NULL |
|    106 | Neha   | 60500.00 |    2021 |       1 | NULL |
|    104 | Sneha  | 60000.00 |    2020 |       2 | NULL |
|    110 | Anjali | 63800.00 |    2020 |       1 | NULL |
|    108 | Pooja  | 65000.00 |    2019 |       3 | NULL |
+--------+--------+----------+---------+---------+------+
10 rows in set (0.00 sec)
```

20. Write a query to rename the DeptName column to DepartmentName.

Answer:

```
mysql> ALTER TABLE DEPT RENAME COLUMN DEPT_NAME TO DEPARTMENTNAME;
Query OK, 0 rows affected (0.05 sec)
Records: 0  Duplicates: 0  Warnings: 0
```